

Matthew Chesnut

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Education

University of Maryland | College Park, MD

- B.S. Mechanical Engineering — Engineering Honors Program (EHP)
- Expected 2027

College of Southern Maryland | LaPlata, MD

- Engineering Coursework
- Dean's List (all semesters)

Experience

Project Engineer Intern | Pillar Construction | Alexandria, VA

May 2026 – September 2026

- Incoming project engineering intern using tools including OST & QBID for estimation & proposal bids. Learning exterior insulation and finish systems (EIFS), sub framing (Knight wall), plaster air barriers (STO PABs), & cement panels (Nichiha panels).
- Learning operations & project management, with field experience. Learning VDC/BIM & site laser scanning & processing the scan data. Finishing the internship with a final focus & capstone presentation.

Stem Tutor | College of Southern Maryland | LaPlata, MD | Prince Frederick, MD

January 2025 – August 2025

- Tutored students in math through calculus III, calculus-based physics, and engineering courses.
- Assisted students in understanding concepts and problem-solving strategies. Built strong connections with students to create a supportive learning environment. Adapted teaching methods to meet individual learning needs.

Auction Operations Team Lead | Bunting Auctions | Owings, MD

April 2022 – August 2025

- Supervised and trained a team of 4 photographers, ensuring efficiency and quality standards for auction listings.
- Assisted customers with pickups, inquiries, and consignor contracts, driving new sign-ups and maintaining strong client relationships.

Projects

- **Geothermal Heat Pump Cycle:** Designed & modeled a dual mode 12.3kW thermal system capable of high efficiency space conditioning (COPs: 6.805 winter / 4.504 summer) using R-454B refrigerant. Performed thermodynamic cycle analysis and component sizing for both summer and winter operating conditions.
- **3-Axis Robotic Arm:** Built and programmed a custom 3-axis robotic arm capable of repeatable, precise object manipulation using stepper motors and embedded motion control.
- **GPS Speedometer:** Built an Arduino-based GPS device to display real-time speed, satellites, and coordinates, programmed in C with a 3D Printed case.

Achievements

- Eagle Scout, Engineering Honors Program

Technical Skills

CAD: SolidWorks, Onshape | **Programming:** MATLAB, Python, C/C++ | **Manufacturing:** MIG Welding, Machining, 3D Printing, Fabrication | **Electronics:** Circuit Design & Analysis, Arduino | **Engineering:** Embedded Systems, Data Analysis, Simulation

